TV Intelligence Scanner — Chipset Report

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Chipsets and memory

ID	Board	Position	Category	Part number	Manufacturer	Qty	Identification	Confidence	Specification (source)	Evidence	Source reference
3	mainboard	U7	Flash memory	W25Q64JVSIQ	Winbond	1	Decoded from vendor part-numbering scheme	75%	density_bits=64 Mbit [decoder]; device_family=SPI flash [marking_rule]; manufacturer=Winbond [decoder]; manufacturer=Winbond [marking_rule]; memory_type=SPI NOR flash [decoder]		decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit); marking rule winbond_spi
2	mainboard	U3	RAM	K4B4G1646E-BYK0	Samsung	2	Datasheet match (exact part number)	90%	data_rate=1600 MT/s [datasheet]; density_bits=4 Gbit [decoder]; density_bits=4 Gbit [datasheet]; device_family=DRAM [marking_rule]; manufacturer=Samsung [decoder]; manufacturer=Samsung [marking_rule]; memory_type=DDR3 SDRAM [decoder]; memory_type=DDR3 [datasheet]; organization=x16 [decoder]; package=96-BALL FBGA [datasheet]; package=FBGA [datasheet]; supply_voltage=1.5 V [datasheet]		datasheet #1 p.1; datasheet #1 p.2; decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation); decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation) (corroborated by datasheet); marking rule samsung_dram

ID	Board	Position	Category	Part number	Manufacturer	Qty	Identification	Confidence	Specification (source)	Evidence	Source reference
1	other		SoC / CPU	MSD6A648	MStar (now MediaTek)	1	Vendor/family from numbering convention only	66%	device_family=TV SoC / scaler [marking_rule]; manufacturer=MStar (now MediaTek) [marking_rule]	2	marking rule mstar_soc

Specification facts

decoder = vendor part-numbering scheme (confirm with datasheet); datasheet = local datasheet text.

Part	Attribute	Value	Source	Reference	Confidence
W25Q64JVSIQ	density_bits	64 Mbit	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%
W25Q64JVSIQ	device_family	SPI flash	marking_rule	marking rule winbond_spi	75%
W25Q64JVSIQ	manufacturer	Winbond	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%
W25Q64JVSIQ	manufacturer	Winbond	marking_rule	marking rule winbond_spi	75%
W25Q64JVSIQ	memory_type	SPI NOR flash	decoder	decoder winbond_spi_nor: Winbond SpiFlash naming (W25Q + density in Mbit)	80%
K4B4G1646E-BYK0	data_rate	1600 MT/s	datasheet	datasheet #1 p.1	60%
K4B4G1646E-BYK0	density_bits	4 Gbit	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation) (corroborated by datasheet)	90%
K4B4G1646E-BYK0	density_bits	4 Gbit	datasheet	datasheet #1 p.1	60%
K4B4G1646E-BYK0	device_family	DRAM	marking_rule	marking rule samsung_dram	70%
K4B4G1646E-BYK0	manufacturer	Samsung	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation)	80%
K4B4G1646E-BYK0	manufacturer	Samsung	marking_rule	marking rule samsung_dram	70%
K4B4G1646E-BYK0	memory_type	DDR3 SDRAM	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation) (corroborated by datasheet)	90%
K4B4G1646E-BYK0	memory_type	DDR3	datasheet	datasheet #1 p.1	65%
K4B4G1646E-BYK0	organization	x16	decoder	decoder samsung_ddr: Samsung DRAM part numbering (K4 + family + density + organisation)	80%
K4B4G1646E-BYK0	package	96-BALL FBGA	datasheet	datasheet #1 p.1	65%

Part	Attribute	Value	Source	Reference	Confidence
K4B4G1646E-BYK0	package	FBGA	datasheet	datasheet #1 p.2	65%
K4B4G1646E-BYK0	supply_voltage	1.5 V	datasheet	datasheet #1 p.1	65%
MSD6A648	device_family	TV SoC / scaler	marking_rule	marking rule mstar_soc	70%
MSD6A648	manufacturer	MStar (now MediaTek)	marking_rule	marking rule mstar_soc	70%